

Comparison of ASME pressure–temperature limits on the fracture probability for a pressurized water reactor pressure vessel



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ABSTRACT

The ASME Code Section XI-Appendix G defines the normal reactor startup (heat-up) and shut-down (cool-down) operation limits according to the fracture toughness requirement of reactor pressure vessel (RPV) materials. This paper investigates the effects of different pressure-temperature limit operations on structural integrity of a Taiwan domestic pressurized water reactor (PWR) pressure vessel. Three kinds of pressure-temperature limits based on different fracture toughness requirements – the K_{Ia} fracture toughness curve of ASME Section XI-Appendix G before 1998 editions, the K_{Ic} fracture toughness curve of ASME Section XI-Appendix G after 2001 editions, and the risk-informed revision method supplemented in ASME Section XI-Appendix G after 2013 editions, respectively, are established as the loading conditions. A series of probabilistic fracture mechanics analyses for the RPV are conducted employing ORNL's FAVOR code considering various radiation embrittlement levels under these pressure-temperature limit conditions. It is found that the pressure-temperature operation limits which provide more operational flexibility may lead to higher fracture risks to the RPV. The cladding-induced shallow surface breaking flaws are the most critical and dominate the fracture probability of the RPV under pressure-temperature limit transients. Present study provides a risk-informed reference for the operation safety and regulation viewpoint of PWRs in Taiwan.

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1. Introduction

The reactor pressure vessel (RPV) is the most critical component in the nuclear power plant (NPP) which has to sustain high temperature, high pressure as well as high radiation environment during operation. Within the reactor beltline region corresponding to the active fuel height, the RPV shell materials frequently become brittle due to the fast neutron irradiation. Considering the degraded fracture toughness of RPV materials, the ASME Code, Section XI, Nonmandatory Appendix G provides a deterministic fracture mechanics methodology for deriving the pressure-temperature limit curves (P-T curves) of normal reactor heat-up (startup) and cool-down (shut-down) operations based on the controlled fracture toughness related to the reference temperature of nil-ductility transition (RT_{NDT}) of beltline region materials. Prior to 1998 editions of ASME Section XI – Appendix G, the required fracture toughness for evaluating P-T limits is the K_{Ia} curve, which is the lower bound on all static, dynamic and arrest fracture toughness (Griesbach, 2006). In 1999, the Code Case N-640 (ASME, 1999)

declared that the fracture toughness requirement can be relaxed from K_{Ia} to K_{Ic} curve, which is the lower bound on static fracture toughness only, and was amended in lieu of K_{Ia} curve in 2001 and the later editions of ASME Section XI – Appendix G. Moreover, to seek both sufficient plant safety and greater operational flexibility, the Electric Power Research Institute (EPRI) proposed a risk-informed revision approach to determine the alternative P-T curves. The demand background, procedures and probabilistic fracture mechanics (PFM) analysis results for the risk-informed methodology were described in MRP-250 report (EPRI, 2009), and this method was also supplemented in 2013 and later editions of ASME Section XI – Appendix G for P-T limit determination. There have been some researchers evaluated the failure risks with respect to such types of P-T curves for RPV by PFM analysis. For instance, Dickson et al. (2010, 2011) have reviewed and re-evaluated in detail about the incremental failure risks of the risk-informed limits on U.S. RPVs, as well as the flaw population effects. Chou and Huang also have compared the failure risks of a Taiwan domestic boiling water reactor (BWR) under K_{Ia} and K_{Ic} based P-T limit operations, respectively (Chou and Huang, 2014).

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In Taiwan, the current P-T curves of the domestic pressurized water reactor (PWR) were established by K_{Ia} curve per 1998 edition of ASME Section XI – Appendix G. Recently, as a result that the

Nomenclature

a_{KIC}	location parameter of Weibull probability function (ksi-in. ^{1/2})	t	wall thickness of RPV beltline region (in.); time (min)
b_{KIC}	scale parameter of Weibull probability function (ksi-in. ^{1/2})	T_C	reactor coolant temperature (°F)
c_{KIC}	shape parameter of Weibull probability function	x	position of flaw tip relative to the inner surface (in.)
c_{pi}	instantaneous conditional probability of initiation	α, β	MRP-250 risk-informed parameters
CR	cool-down rate (°F/h)	τ	time step of each transient (min)
f	neutron fluence (10 ¹⁹ n/cm ²)	BWR	boiling water reactor
f_{surf}	neutron fluence at inner wetted surface (10 ¹⁹ n/cm ²)	CPI	conditional probability of initiation
f_{P-T}	frequency for operation on or near the P-T limits (/year)	CPF	conditional probability of failure
h	convective heat-transfer coefficient (Btu/h-ft ² -°F)	EPFY	effective full power year
HR	heat-up rate (°F/h)	EPRI	Electric Power Research Institute
K_I	applied stress intensity factor (ksi-in. ^{1/2})	FAVOR	Fracture Analysis of Vessels-Oak Ridge
K_{Ia}, K_{IC}	required fracture toughness (ksi-in. ^{1/2})	IGA	initiation-growth-arrest
K_{Im}	applied stress intensity factor by membrane stress (ksi-in. ^{1/2})	NPP	Nuclear power plant
K_{IT}	applied stress intensity factor by thermal stress (ksi-in. ^{1/2})	ORNL	Oak Ridge National Laboratory
M_m	coefficient for determining K_{Im} from membrane stress (in. ^{1/2})	PFM	probabilistic fracture mechanics
P	permissible pressure (ksi)	PNNL	Pacific Northwest National Laboratory
R_i	inner radius of RPV (in.)	P-T	pressure-temperature
RT_{NDT}	reference temperature of nil-ductility transition (°F)	PTS	pressurized thermal shock
T	crack tip temperature (°F)	PWR	pressurized water reactor
		RPV	reactor pressure vessel
		SIFIC	stress intensity factor influence coefficient
		TWCF	through-wall cracking frequency
		WPS	warm-prestressing

latest surveillance capsule (Y Capsule) of the PWR vessel has been taken out and tested, the licensee proposed the new P-T curves based on the K_{IC} curve per 2007 edition of ASME Section XI – Appendix G. Hence, a series of evaluations need to be performed first to ensure that the domestic PWR pressure vessel can maintain acceptable fracture risk when different P-T curves are applied. In this paper, the fracture risks of the domestic PWR vessel under P-T limit transients calculated by different methodologies are compared. We considered three types of P-T curves based on K_{Ia} curve, K_{IC} curve and the risk-informed revision method, respectively. At first, these types of P-T curves for the domestic PWR at multiple levels of radiation embrittlement and different heat-up or cool-down rates were established as the loading conditions. Then, a series of PFM analyses were conducted to compare the failure risks under the various P-T limit operations. In this work, the FAVOR (Fracture Analysis of Vessels-Oak Ridge) code developed by the Oak Ridge National Laboratory (ORNL) in the United States (Williams et al., 2012; Dickson et al., 2012) was employed for PFM analysis. The model consistent with the previous pressurized thermal shock (PTS) evaluation for the domestic PWR pressure vessel (Chou and Huang, 2016) was utilized. In addition, some deterministic analyses were also performed to verify the trends of the PFM analysis results. Present study can be a reference for the regulatory body to review the operational limits of the domestic nuclear power plants in Taiwan.

2. ASME P-T limit transients

ASME Code Section XI – Appendix G specifies the deterministic fracture methodology for deriving the maximum permissible pressure for normal heat-up transients associated with reactor startup and normal cool-down transients associated with reactor shut-down, which follows (Chou and Huang, 2014; Dickson et al., 2010):

- (1) A surface flaw with one-fourth depth of the reactor vessel thickness ($t/4$) as well as 6:1 aspect ratio is postulated to calculate the stress intensity factor histories under the normal

cool-down and heat-up transients. The reference flaw is postulated to reside in the region with the highest value of RT_{NDT} .

- (2) Considering a safety factor of 2 on pressure stress, the sum of the stress intensity factors due to pressure stress and thermal stress should be less than the required fracture toughness, giving:

$$2K_{Im} + K_{IT} < K_{Ia} \text{ or } K_{IC} \quad (1)$$

The time-dependent fracture toughness requirements, $K_{Ia}(t)$ for 1998 or earlier editions of ASME Section XI-Appendix G, or $K_{IC}(t)$ for 2001 or later editions of ASME Section XI-Appendix G, are described by the following equations (ksi-in.^{1/2}):

$$K_{Ia}(t) = 26.78 + 1.223 \exp[0.0145(T(t) - RT_{NDT} + 160)] \quad (2)$$

$$K_{IC}(t) = 33.2 + 20.734 \exp[0.02(T(t) - RT_{NDT})] \quad (3)$$

Herein the time-dependent crack tip temperature, $T(t)$, should be derived from the permissible coolant temperature $T_C(t)$ according to the prescribed heat-up and cool-down rate.

- (3) In Eq. (1), $K_{Im}(t) = M_m(PR_i/t)$ is the time-dependent applied stress intensity factor produced by pressure-induced membrane loading in the RPV shell, where
 $P(t)$ = time-dependent internal pressure (ksi),
 R_i = internal radius of RPV (in.),
 t = vessel wall thickness (in.),

And M_m is specified in ASME Code as follows:

$M_m = 0.893\sqrt{t}$ for axially oriented external-surface breaking reference flaw,
 $M_m = 0.443\sqrt{t}$ for circumferentially oriented external-surface breaking reference flaw,
 $M_m = 0.926\sqrt{t}$ for axially oriented internal-surface breaking reference flaw,

$M_m = 0.443\sqrt{t}$ for circumferentially oriented internal-surface breaking reference flaw.

The other term, K_{II} , in Eq. (1) is the thermal stress intensity factors calculated based on radial thermal gradients from heat-up and cool-down conditions which can be simply determined as follows:

For an axially or circumferentially oriented inside surface defect:

$$K_{II} = 0.953 \times 10^{-3} \times CR \times t^{2.5} \quad (4)$$

For an axially or circumferentially oriented outside surface defect:

$$K_{II} = 0.753 \times 10^{-3} \times HR \times t^{2.5} \quad (5)$$

where CR is the cool-down rate (°F/h) and HR is the heat-up rate (°F/h), respectively. Therefore, the maximum permissible coolant pressure is derived as follows:

$$P(t) = [\{K_{Ia}(t) \text{ or } K_{IC}(t)\} - K_{II}] \cdot t/R_i/2M_m \quad (6)$$

For reactor normal cool-down transients, Eq. (6) is evaluated at the deepest point of the internal-surface breaking reference flaw whereas for heat-up transients, the maximum allowable coolant pressure-temperature relationship is the minimum pressure at any coolant temperature determined from:

- (1) Application of Eq. (6) evaluated at the deepest point of the outside reference flaw.
- (2) Application of Eq. (6) considering the steady state solution ($K_{II}(t)=0$) evaluated at the deepest point of both the internal-surface breaking reference flaw and external-surface breaking reference flaw.

3. Risk-informed P-T limit transients

MRP-250 report (EPRI, 2009) proposed the risk-informed parameters, α and β , to determine permissible pressure under heat-up and cool-down at rates up to 100 °F/h. Substituting the expression for $K_{IC}(t)$ into Eq. (6), the maximum permissible coolant pressure can be derived as the generalized form:

$$P(t) = \{33.2 + 20.734 \exp[0.02(T(t) - RT_{NDT} - \beta)] - K_{II}\} \times (t/R_i) \times (1/\alpha) \times (1/M_m) \quad (7)$$

For the methodology of current ASME code, $\alpha = 2$ and $\beta = 0$ are used in Eq. (7) to derive P-T curves. The value of $\alpha = 2$ means the safety factor used to accommodate additional sources of stress intensity factor other than the thermal gradient, including the weld residual stresses, compressive stresses on the crack surface, and cladding stresses (Dickson et al., 2010). For MRP-250 risk-informed methodology, $\alpha = 1$ and $\beta = 110$ °F was finally suggested and also upgraded in 2013 and later editions of ASME Section XI – Appendix G. Therefore, the risk-informed allowable pressure of the ASME code can be shown below:

$$P(t) = \{33.2 + 20.734 \exp[0.02(T(t) - RT_{NDT} - 110)] - K_{II}\} \times (t/R_i) \times (1/M_m) \quad (8)$$

Fig. 1 illustrates the comparison of P-T curves derived by different methodologies in this work. As seen from the figure, the K_{Ia} curve method in earlier ASME code produces the lowest limit. Between the P-T curves derived by K_{IC} curve method and risk-informed method in the current ASME code, the value of α has a substantial influence on the pressure limit in the lower temperature, while the value of β largely affects the time or temperature at which the operating pressure reaches its maximum allowable value in the high temperature region (EPRI, 2009).

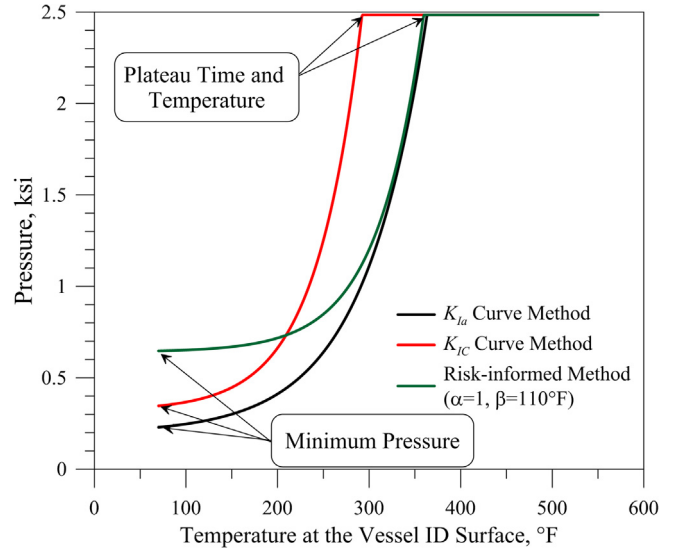


Fig. 1. Different types of P-T curves.

For cool-down transients, introducing $\beta = 110$ °F shifts the time at which the pressure may remain at the steady state operating pressure before being reduced, to an earlier time in the transient. Reducing $\alpha = 2-1$ raises the allowable coolant pressure late in the transient, which improves the operational flexibility. On the contrary, for heat-up transients, the primary impact of introducing $\beta = 110$ °F is to delay the plateau time and temperature that takes for the pressure to reach the steady state operating pressure, to a later time in the transient. Reducing α from 2 to 1 raises the allowable coolant pressure early in transient, again, which improves the operational flexibility (Dickson et al., 2010, 2011).

4. PFM analysis method

The PFM FAVOR code version 12.1 was employed to compute the fracture risks of the domestic PWR pressure vessel under various types of P-T limit transients. Two computational modules of the FAVOR were used in the study: (1) the deterministic load generator (FAVLoad), and (2) the Monte Carlo PFM module (FAVPFM). The input file for FAVLoad contains the material properties, RPV geometry, and thermal hydraulic conditions of transients. At first FAVLoad calculates and subsequently outputs the histories of temperature, stress and stress intensity factor, K_I , of crack tips with various depth and aspect ratio along the wall thickness of the RPV under each transient. Based on the output of FAVLoad, the PFM module, FAVPFM, calculates the instantaneous conditional probability of initiation (cpi) of each simulated flaw tip at each time step, τ , during the transient by the Weibull probability function, giving (Williams et al., 2012):

$$cpi(\tau) = \begin{cases} 0, & K_I(\tau) \leq a_{KIC} \\ 1 - \exp\left\{-\left[\frac{K_I(\tau) - a_{KIC}}{b_{KIC}}\right]^{c_{KIC}}\right\}, & K_I(\tau) > a_{KIC} \end{cases} \quad (9)$$

In Eq. (9), a_{KIC} , b_{KIC} and c_{KIC} denote the location parameter, scale parameter and shape parameter of the random variate, respectively. They are functions of $T-RT_{NDT}$ as follows:

$$\begin{aligned} a_{KIC}(T - RT_{NDT}) &= 19.35 + 8.335 \exp[0.02254 \cdot (T - RT_{NDT})] \\ b_{KIC}(T - RT_{NDT}) &= 15.61 + 50.132 \exp[0.008 \cdot (T - RT_{NDT})] \\ c_{KIC} &= 4 \end{aligned}$$

FAVOR defines the conditional probability of initiation (CPI) of each simulated flaw during the entire transient as the maximum value of $cpi(\tau)$. The warm-prestressing (WPS) effect, which is

characterized as an apparent increase of fracture toughness of ferritic steels after first being “prestressed” at an elevated temperature, was also taken into account. Therefore, most crack initiation especially under the cool-down P-T limit transients may be excluded due to the WPS effect, which reduces the predicted fracture probability (Chou and Huang, 2014, 2016; Qian and Niffenegger, 2014). In the study, the baseline WPS model of FAVOR was applied in all of the PFM analyses. If a flaw is in a state of WPS, it is not eligible for initiation (or re-initiation if it has arrested) until following conditions are met (Williams et al., 2012):

- (1) The applied- K_I of the flaw tip is great than the lower bound of fracture toughness, a_{KIC} , in Eq. (9) for the temperature at the flaw tip (applied- $K_I(\tau) > a_{KIC}(\tau)$).
- (2) A raising applied- K_I field, i.e. the time-rate-of-change of the applied- K_I is positive ($dK_I(\tau)/d\tau > 0$).
- (3) In a rising applied- K_I field, the applied- K_I must be larger than the previously-established maximum value experienced by the flaw tip during the transient up to the current point in time under consideration (applied- $K_I(\tau) > K_{I(\max)}(\tau)$).

When a crack initiates, it is assumed to become surface breaking with infinite aspect ratio based on the results observed from the large-scale fracture experiments which showed that the initiated flaws in cleavage fracture was to extend in length before propagating through the wall thickness (Williams et al., 2012). Then, the Initiation-Growth-Arrest (IGA) submodel will calculate the proportion of subsequent through wall cracking considering the crack propagation, plastic collapse and ductile-tearing behavior. The conditional probability of failure (CPF) of each simulated flaw is defined as the crack propagation up to 90% of vessel wall thickness rather than the through-wall cracking.

5. Analysis model

5.1. PFM model for RPV beltline region

Fig. 2 illustrates the beltline region configuration of the domestic PWR pressure vessel. The inner radius and wall thickness of the

PWR pressure vessel beltline region are 78.5 in. and 8.04 in. which includes 0.16 in. stainless-steel cladding, respectively. Table 1 lists the temperature dependent thermo-mechanical properties of the base metal and cladding material of the RPV. The base and cladding stress free temperature of 488 °F recommended by FAVOR's theory consistent with the PTS re-evaluation study (Williams et al., 2012) was applied. For FAVOR probabilistic analysis, there are 9 major regions divided in the beltline region model. Major regions 1–4 are the axial welds 101-124A/B and 101-142A/B; major region 5 is the circumferential weld 101-171; for other 4 regions are plates. Their corresponding sizes within the beltline region are also noted in Fig. 2. In the analysis, the adjusted RT_{NDT} caused by neutron irradiation, was calculated by the Eason's correlation (Eason et al., 2007), which is consistent with the U.S. NRC's PTS re-evaluation (EricksonKirk and Dickson, 2010) and MRP-250 report (EPRI, 2009). Compared with the conventional correlation in Regulatory Guide 1.99 Rev. 2 (U.S. NRC, 1988), the new correlation was developed from a wider research database and with more comprehensive embrittlement mechanisms. The new correlation not only considers the variables copper, nickel, and neutron fluence that were in Regulatory Guide 1.99 Rev. 2, but also includes neutron flux level, irradiation temperature, phosphorus and manganese contents, effective copper parameter, etc. Besides, the margin term in Regulatory Guide 1.99 Rev. 2 considering the uncertainty is also omitted. Therefore, the revised correlation is much more complicated but realistic. The neutron fluence (10^{19} n/cm², $E > 1$ MeV), f , which attenuates along the vessel wall, can be evaluated as below:

$$f = f_{\text{surf}}(e^{-0.24x}) \quad (10)$$

Above, f_{surf} is the neutron fluence at the inner wetted surface and x is the position of the flaw tip (in inches) from the inner surface. Table 2 lists the weight percentage of the decisive chemical contents which include copper, nickel, manganese and phosphorous, and the initial RT_{NDT} of each beltline region material provided by the manufacturer (Li et al., 2014; Singer, 1982). According to the FAVOR's theory, all embedded weld flaws are allocated along the fusion line area which means the boundaries between a weld and its adjacent plates. The corresponding RT_{NDT} at each flaw tip

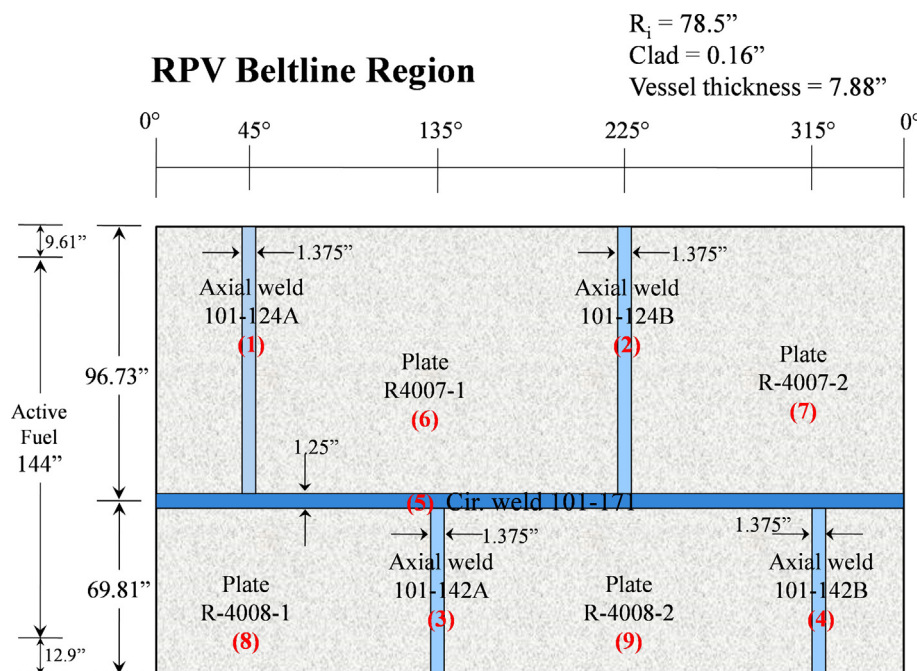


Fig. 2. The configuration of welds and plates in the RPV beltline region.

Table 1
Thermo-mechanical properties of RPV materials.

Cladding																
Temperature (°F)	70	100	150	200	250	300	350	400	450	500	550	600	650	700	750	800
Young's modulus (ksi)	22,045	21,799	21,389	20,979	20,569	20,159	19,719	19,280	18,840	18,400	17,960	17,521	17,081	16,641	16,201	15,762
Poisson's ratio	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Thermal expansion coefficient ($10^{-6} \text{ } ^\circ\text{F}^{-1}$)	8.48	8.55	8.67	8.79	8.9	9	9.1	9.19	9.28	9.37	9.45	9.53	9.61	9.69	9.76	9.82
Thermal conductivity (BTU/h-ft-°F)	8.1	8.4	8.6	8.8	9.1	9.4	9.6	9.9	10.1	10.4	10.6	10.9	11.1	11.4	11.6	11.9
Specific heat (BTU/lbm-°F)	0.116	0.119	0.12	0.121	0.123	0.126	0.126	0.128	0.129	0.131	0.131	0.133	0.134	0.135	0.136	0.137
Density (lbm/ft ³)	489	489	489	489	489	489	489	489	489	489	489	489	489	489	489	489
Base Metal																
Temperature (°F)	70	100	150	200	250	300	350	400	450	500	550	600	650	700	750	800
Young's modulus (ksi)	29,200	29,039	28,769	28,500	28,250	28,000	28,370	27,400	27,200	27,000	26,700	26,400	28,850	25,300	24,600	23,900
Poisson's ratio	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Thermal expansion coefficient ($10^{-6} \text{ } ^\circ\text{F}^{-1}$)	7.00	7.06	7.16	7.25	7.34	7.43	7.5	7.58	7.63	7.7	7.77	7.83	7.9	7.94	8	8.05
Thermal conductivity (BTU/h-ft-°F)	24.8	25	25.1	25.2	25.2	25.1	25	25.1	24.6	24.3	24	23.7	23.4	23	22.6	22.2
Specific heat (BTU/lbm-°F)	0.105	0.107	0.11	0.114	0.117	0.119	0.122	0.127	0.128	0.13	0.133	0.135	0.138	0.14	0.144	0.147
Density (lbm/ft ³)	489	489	489	489	489	489	489	489	489	489	489	489	489	489	489	489

Table 2
Embrittlement-related properties of the beltline region materials.

Material ID	Major Region	Copper Content (wt.%)	Nickel Content (wt.%)	Phosphorous Content (wt.%)	Manganese Content (wt.%)	Initial RT_{NDT} (°F)
Axial welds	101-124A	1	0.03	0.1	0.008	1.25
	101-124B	2	0.06	0.11	0.006	1.34
	101-142A	3	0.02	0.11	0.002	1.19
	101-142B	4	0.02	0.11	0.002	1.25
Circumferential weld	101-171	5	0.042	0.129	0.017	1.39
Plates	R4007-1	6	0.063	0.613	0.006	1.34
	R4007-2	7	0.064	0.597	0.005	1.45
	R4008-1	8	0.053	0.620	0.01	1.54
	R4008-2	9	0.07	0.670	0.007	1.36

is the highest one sampled between the weld and its adjacent plates (Williams et al., 2012).

Four levels of radiation embrittlement were considered. Two neutron fluence maps corresponding to the present end-of-license operation (36 effective full power years (EFPY)) and the proposed extended operation (54 EFPY) were constructed according to the analysis results for the latest surveillance capsule (Y Capsule) (Li et al., 2014). Two additional levels considerably beyond the operational lifetimes (200 EFPY and 500 EFPY) extrapolated by the current neutron fluence map were also considered. The maximum neutron fluence values on each major region corresponding to various embrittlement levels are listed in Table 3. The beltline region model which consists of various embrittlement-relative properties and neutron fluence values were written in the input file of FAVPFM.

For establishing the P-T curves, each beltline region material was assumed to be subjected to the maximum neutron fluence values of $5.50 \times 10^{19} \text{ n/cm}^2$, $8.25 \times 10^{19} \text{ n/cm}^2$, $30.55 \times 10^{19} \text{ n/cm}^2$ and $76.39 \times 10^{19} \text{ n/cm}^2$ for 36, 54, 200 and 500 EFPY, respectively. The highest projected RT_{NDT} in the PWR pressure vessel beltline region at various embrittlement levels are listed in Table 4, which were used to calculate the P-T curves. Figs. 3 and 4 present the allowable P-T limit transients of the RPV at a rate of $100 \text{ } ^\circ\text{F/h}$ for cool-down and heat-up respectively based on different methodologies. It is seen that the permissible pressure decreases significantly with the increased radiation embrittlement. The risk-informed P-T curves have the higher permissible pressure in the lower temperature region, i.e. the late stage for cool-down transients and the early stage for heat-up transients. The transition in the heat-up limits for higher embrittlement levels that reduces the permissible pressure

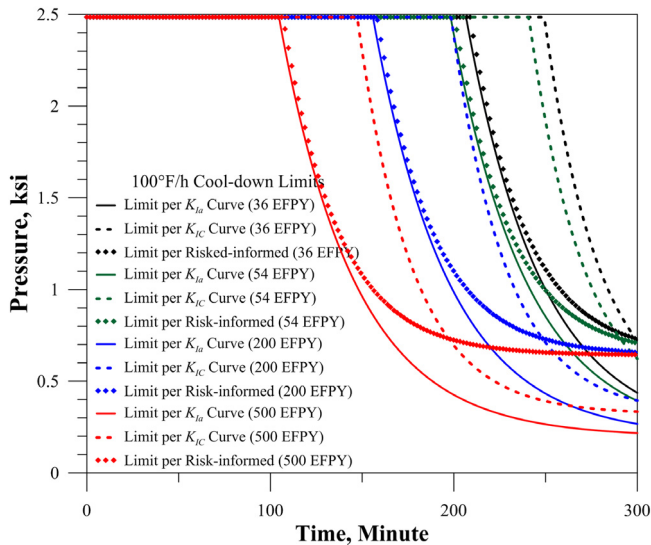
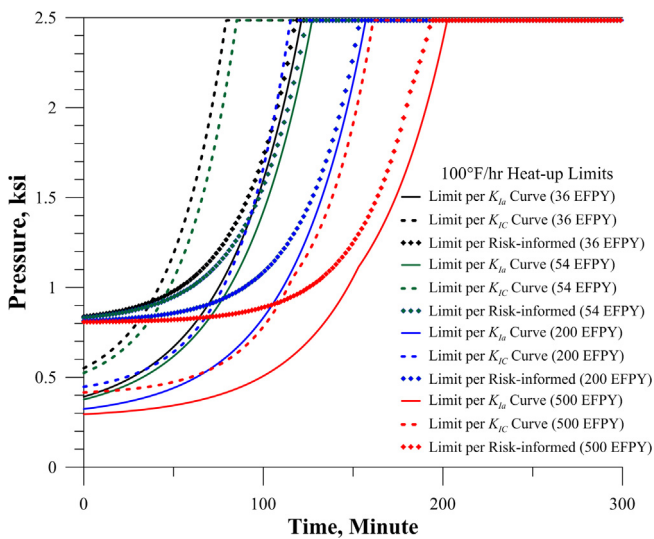
Table 3
The maximum neutron fluences of beltline major regions.

Material ID		Major Region	Ratio to Max. neutron flux	Neutron fluence ^a (10 ¹⁹ n/cm ²)			
				36 EFPY	54 EFPY	200 EFPY	500 EFPY
Axial welds	101-124A	1	30	1.65	2.48	9.17	22.92
	101-124B	2	30	1.65	2.48	9.17	22.92
	101-142A	3	30	1.65	2.48	9.17	22.92
	101-142B	4	30	1.65	2.48	9.17	22.92
Circumferential weld	101-171	5	100	5.50	8.25	30.55	76.39
Plates	R4007-1	6	99	5.45	8.17	30.24	75.63
	R4007-2	7	99	5.45	8.17	30.24	75.63
	R4008-1	8	100	5.50	8.25	30.55	76.39
	R4008-2	9	100	5.50	8.25	30.55	76.39

^a The maximum values of the beltline region are $5.50 \times 10^{19} \text{ n/cm}^2$, $8.25 \times 10^{19} \text{ n/cm}^2$, $30.55 \times 10^{19} \text{ n/cm}^2$ and $76.39 \times 10^{19} \text{ n/cm}^2$ for 36, 54, 200 and 500 EFPY, respectively.

Table 4The highest projected RT_{NDT} of the RPV beltline region.

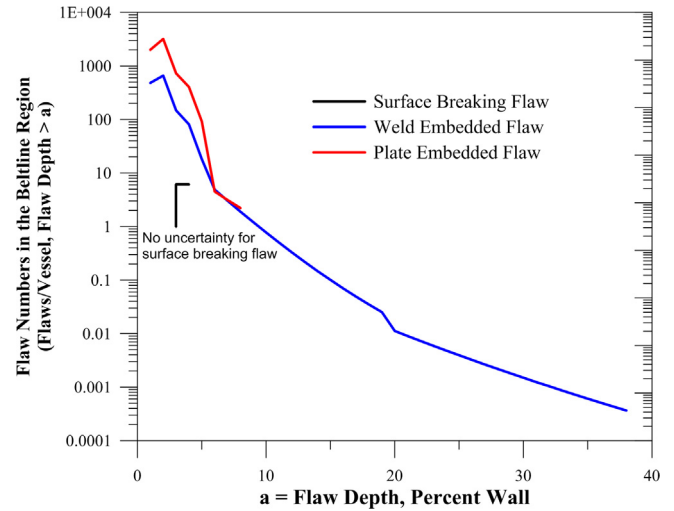
Embrittlement levels	RT_{NDT} (°F)		
	Clad-base interface	1/4 T inner crack	3/4 T outer crack
36 EFPY	85.2	72.3	53.9
54 EFPY	102.1	86.3	63.8
200 EFPY	187.2	156.9	113.5
500 EFPY	290.2	242.2	173.7

**Fig. 3.** Various cool-down P-T limit transients as a rate of 100 °F/h.**Fig. 4.** Various heat-up P-T limit transients as a rate of 100 °F/h.

after a while denotes the steady state control by the solution evaluated at the internal-surface breaking reference flaw.

With regard to the PFM analysis, some modeling assumptions were made for conservatism:

- (1) Each major region is subjected to its maximum neutron fluence value listed in Table 3. Practically, the neutron fluence varies along the circumferential and vertical directions on each major region.

**Fig. 5.** Mean flaw number distribution of the RPV beltline region against the percentage of the vessel wall thickness.

- (2) Some additional requirements of 10 CFR 50, Appendix G (U.S. NRC, 2008) which restrict the permissible pressure were not considered. For example, the prescribed pressure should be less than 20% of preservice system hydrostatic test pressure when the coolant temperature is lower than the highest RT_{NDT} of closure flange region plus 120 °F.
- (3) The convective heat-transfer coefficient was conservatively set to a constant value of $h = 10,000$ Btu/h-ft²-°F since this is considered to be conduction limited, i.e. the condition in which the wall conduction dominates the thermal resistance to energy flowing from the wall to the fluid (Dickson et al., 2010).
- (4) The RPV inner surface cladding was assumed single layer that the surface breaking flaws were postulated to exist. It has been demonstrated that the surface breaking flaws physically exist in RPV steels due to clad fabrication defects (Dickson et al., 2011).

Consequently, there were 72 PFM analyses in total performed in the study: 6 (4 cool-down rates: 20, 40 60 and 100 °F/h; 2 heat-up rates: 60 and 100 °F/h) $\times$ 4 (radiation embrittlement levels: 36, 54, 200 and 500 EFPY) $\times$ 3 (method of deriving permissible pressure: K_{Ia} curve method, K_{IC} curve method and risk-informed method) = 72 cases.

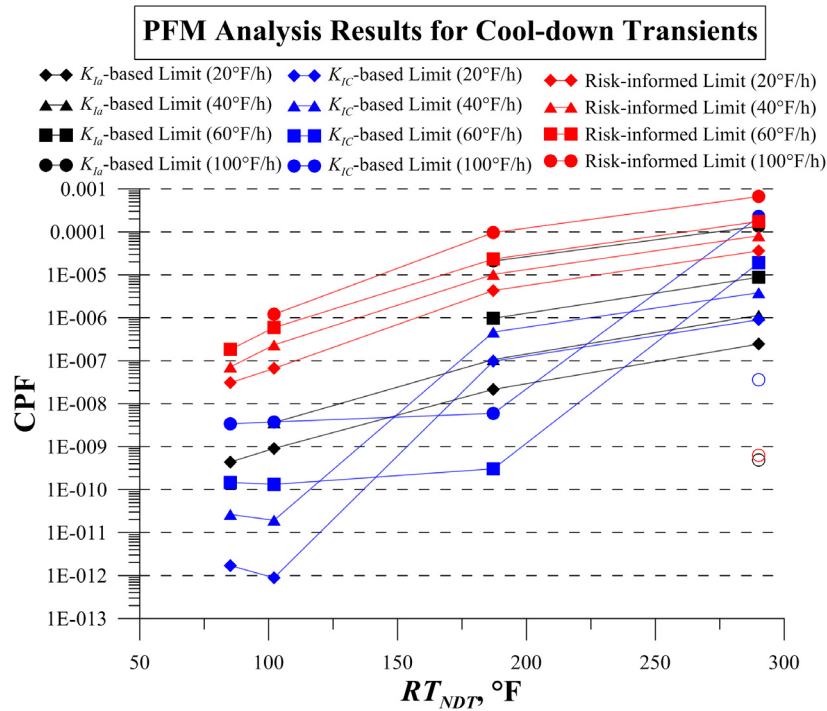
5.2. Flaw characterizations

Three flaw files involving the flaw characteristics of surface breaking flaw, S.dat, embedded weld flaw, W.dat, and embedded plate flaw, P.dat, for FAVOR were generated by PNNL's VFLAW code (Simonen et al., 2003) based on a series of researches and measurements for non-destructive and destructive examination of actual vessel materials. The flaw characteristic is based on the welding process, material, RPV geometry, etc. of the RPV. Therefore the relevant parameters should be input for VFLAW to generate the flaw files for FAVOR. FAVOR assumes all pre-existing internal surface breaking flaws simulated from S.dat are circumferentially oriented. This assumption is on the basis of the vessel fabrication, which has austenitic stainless-steel cladding applied to the inner surface of the vessel. There is no uncertainty with respect to the number of flaws or the depth for the postulated surface breaking flaws. The aspect ratio of surface breaking flaws for the domestic RPV is distributed as follows: 57.82% have aspect ratio of 2, 28.88% have

Table 5

The PFM results of the domestic PWR vessel under P-T limit transients.

P-T limits	Rate (°F/h)	Method	36 EFPY		54 EFPY		200 EFPY		500 EFPY	
			CPI	CPF	CPI	CPF	CPI	CPF	CPI	CPF
Cool-down	20	K_{Ia} Curve	1.6×10^{-8}	4.4×10^{-10}	4.3×10^{-8}	9.1×10^{-10}	9.0×10^{-7}	2.2×10^{-8}	5.0×10^{-6}	2.5×10^{-7}
		K_{IC} Curve	4.5×10^{-12}	1.7×10^{-12}	2.6×10^{-12}	8.8×10^{-13}	1.3×10^{-6}	9.7×10^{-8}	7.2×10^{-6}	9.0×10^{-7}
		Risk-informed	1.2×10^{-7}	3.1×10^{-8}	3.2×10^{-7}	6.6×10^{-8}	7.5×10^{-6}	4.3×10^{-6}	4.8×10^{-5}	3.6×10^{-5}
	40	K_{Ia} Curve	0	0	1.1×10^{-7}	3.6×10^{-9}	2.0×10^{-6}	1.1×10^{-7}	1.1×10^{-5}	1.1×10^{-6}
		K_{IC} Curve	6.3×10^{-11}	2.6×10^{-11}	5.2×10^{-11}	1.9×10^{-11}	2.9×10^{-6}	4.7×10^{-7}	1.6×10^{-5}	3.9×10^{-6}
		Risk-informed	2.1×10^{-7}	7.2×10^{-8}	5.8×10^{-7}	2.3×10^{-7}	1.5×10^{-5}	1.0×10^{-5}	9.8×10^{-5}	8.1×10^{-5}
	60	K_{Ia} Curve	0	0	0	0	4.6×10^{-6}	9.7×10^{-7}	2.6×10^{-5}	8.8×10^{-6}
		K_{IC} Curve	3.2×10^{-10}	1.5×10^{-10}	3.1×10^{-10}	1.3×10^{-10}	6.2×10^{-10}	3.1×10^{-10}	4.0×10^{-5}	1.9×10^{-5}
		Risk-informed	4.0×10^{-7}	1.8×10^{-7}	1.1×10^{-6}	5.9×10^{-7}	3.0×10^{-5}	2.3×10^{-5}	2.0×10^{-4}	1.8×10^{-4}
	100	K_{Ia} Curve	0	0	0	0	3.2×10^{-5}	2.1×10^{-5}	1.9×10^{-4}	1.4×10^{-4}
		K_{IC} Curve	5.2×10^{-9}	3.4×10^{-9}	6.0×10^{-9}	3.8×10^{-9}	1.1×10^{-8}	6.0×10^{-9}	2.8×10^{-4}	2.3×10^{-4}
		Risk-informed	0	0	1.8×10^{-6}	1.2×10^{-6}	1.1×10^{-4}	9.6×10^{-5}	7.2×10^{-4}	6.7×10^{-4}
Heat-up	60	K_{Ia} Curve	2.1×10^{-9}	5.9×10^{-13}	7.1×10^{-9}	4.3×10^{-12}	2.4×10^{-7}	4.5×10^{-10}	1.4×10^{-6}	6.3×10^{-9}
		K_{IC} Curve	5.8×10^{-9}	1.0×10^{-10}	1.7×10^{-8}	2.5×10^{-10}	4.2×10^{-7}	7.2×10^{-9}	2.3×10^{-6}	7.0×10^{-8}
		Risk-informed	2.6×10^{-8}	3.0×10^{-9}	7.8×10^{-8}	1.0×10^{-8}	2.0×10^{-6}	5.6×10^{-7}	1.2×10^{-5}	5.4×10^{-6}
	100	K_{Ia} Curve	8.4×10^{-10}	0	3.2×10^{-9}	0	1.3×10^{-7}	8.9×10^{-12}	8.3×10^{-7}	1.3×10^{-10}
		K_{IC} Curve	2.3×10^{-9}	7.2×10^{-13}	7.5×10^{-9}	4.6×10^{-12}	2.4×10^{-7}	4.7×10^{-10}	1.5×10^{-6}	8.1×10^{-9}
		Risk-informed	1.0×10^{-8}	4.3×10^{-10}	3.3×10^{-8}	1.7×10^{-9}	1.0×10^{-6}	1.2×10^{-7}	6.2×10^{-6}	1.5×10^{-6}

**Fig. 6.** The CPF variations against the maximum beltline surface RT_{NDT} for cool-down P-T limit transients.

aspect ratio of 6, 3.92% have aspect ratio of 10, and 9.38% have aspect ratio of infinity.

Embedded weld flaws are oriented axially for axial welds, and oriented circumferentially for circumferentially welds. For embedded flaws in plates, 50% of both axially and circumferentially oriented flaws are assigned. Uncertainties with respect to flaw number, depth and aspect ratio for embedded flaws are simulated by VFLAW. Fig. 5 shows the mean flaw numbers of the PWR pressure vessel beltline region against the percentage of the vessel wall thickness based on 1000 Monte Carlo simulations by VFLAW.

Further, the Flaw Population Option 3 of FAVOR, which allocates the surface breaking and embedded flaws along both inner and outer vessel wall, was used to determine the potential effects of heat-up transients on external flaws of RPV.

5.3. Acceptable fracture risk

In the paper, the acceptable fracture risk for the domestic RPV under the P-T limit operations refers to the U.S. NRC's new PTS through-wall cracking frequency (TWCF) limit of 1×10^{-6} /year (EricksonKirk et al., 2007; EricksonKirk and Dickson, 2010; U.S. NRC, 2010). The TWCF can be expressed as:

$$TWCF = CPF \times F_{P-T} < 1 \times 10^{-6}/\text{year} \quad (11)$$

where CPF is the condition probability of vessel failure due to through wall cracking under the operation on or near the P-T limits, and F_{P-T} is the corresponding event frequency. In practice, the reactor normal heat-up and cool-down operations are within the procedural and design system constraints, which preclude operation near

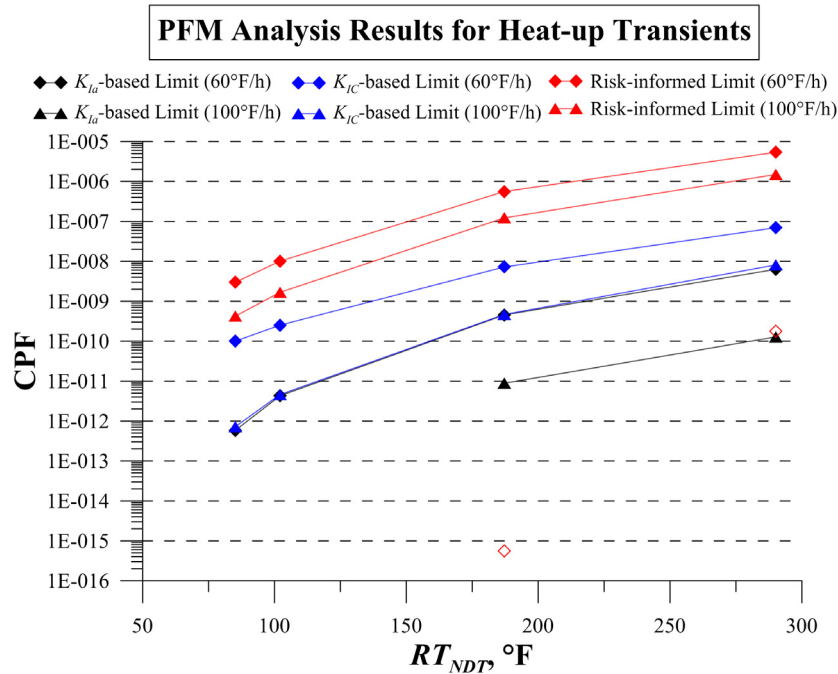


Fig. 7. The CPF variations against the maximum beltline surface RT_{NDT} for heat-up P-T limit transients.

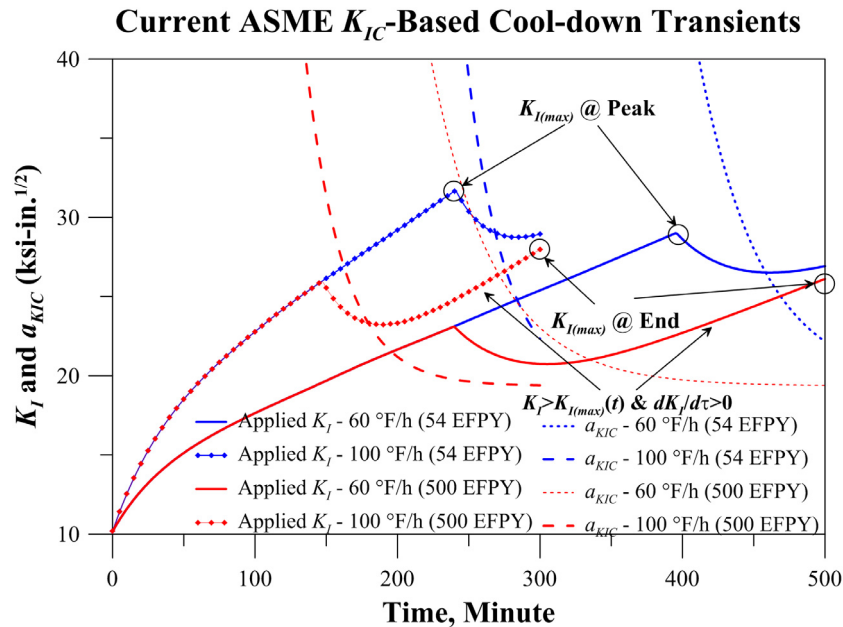


Fig. 8. The deterministic analysis results for a postulated flaw under K_{IC} -based cool-down P-T limit transients.

the P-T limits. For events with inadvertent operation up to P-T limits, service experience indicates that F_{P-T} is less than approximately 0.01/year (EPRI, 2009). Here, F_{P-T} is conservatively assumed to be 0.1/year for the domestic PWR pressure vessel. Therefore, the calculated CPF less than 1×10^{-5} for the domestic RPV would be acceptable to meet the safety goal.

6. Results and discussion

Table 5 lists the conditional probabilities of crack initiation (CPI) and crack penetration (CPF) computed by FAVOR. Also, the

CPF as a function of the maximum surface RT_{NDT} of the domestic RPV beltline region under cool-down and heat-up P-T limit transients, respectively are shown in Figs. 6 and 7. Each case was obtained by performing 100,000 Monte Carlo simulations considering both the calculation time and convergency of analysis results. The hollow symbol in the figures denotes the CPFs contributed by the embedded flaws. The PFM analysis results indicate that the highest CPF is around 6.7×10^{-4} calculated from the 500 EFPY, 100 °F/h cool-down rate P-T limit transient per risk-informed method. For embrittlement levels within 54 EFPY, all P-T curves derived from different methods can meet the safety goal ($CPF < 1 \times 10^{-5}$). Therefore, it can be said that the domestic PWR

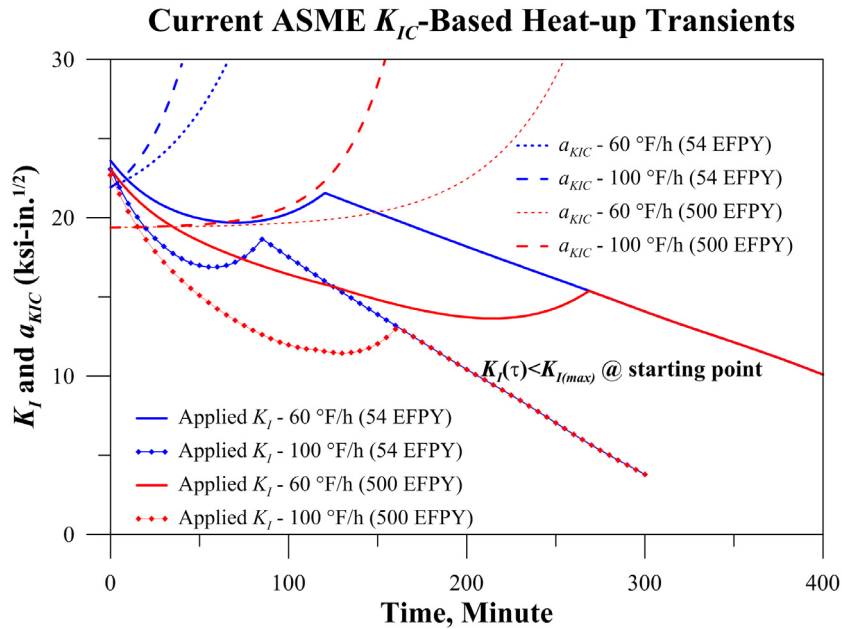


Fig. 9. The deterministic analysis results for a postulated flaw under K_{IC} -based heat-up P-T limit transients.

pressure vessel has sufficient structural integrity if its original P-T limits per earlier ASME (K_{Ia} curve) are updated to the current ASME (K_{IC} curve) limits, even to the risk-informed limits which have more operational flexibility.

According to the analysis results, some trends are found in general:

- (1) The fracture risk of the RPV under P-T limit transient increases significantly with the increased RT_{NDT} due to the reduction of fracture toughness, except for some K_{IC} -based cool-down limits between 36 EFPY and 54 EFPY. The reason for the exception can be attributed to that the effects of permissible pressure decreased by the increased RT_{NDT} may be more dominant than the fracture toughness reduction on fracture probability of the RPV.
- (2) Instead of numerous embedded flaws, the cladding-induced shallow surface breaking flaws dominate the fracture behavior of the RPV. Such phenomenon is consistent with previous researches (Dickson et al., 2010, 2011). Only very tiny proportions of failure risk are computed by embedded flaws under P-T limits at higher levels of radiation embrittlement.
- (3) For both cool-down and heat-up transients, risk-informed limits produce the highest failure risks than the K_{IC} -based limits, and the K_{Ia} -based limits of earlier ASME are the least.

However, there are also some discrepancies observed from the PFM analysis results. Therefore, we have also performed a series of deterministic analyses by FAVOR to explain the exceptional results. Because the cladding-induced shallow surface breaking flaws dominate the fracture behavior, the postulated flaw for FAVOR deterministic analysis was considered to be a circumferential inner surface breaking flaw with the depth of 2% the RPV thickness (cladding thickness) and the aspect ratio of 10. On the basis of deterministic analysis results, the discrepancies and their corresponding reasons are discussed as follows:

- (1) The fracture probability increases with increasing the cool-down rate, but decreases with increasing the heat-up rate. Figs. 8 and 9 present the FAVOR deterministic analysis

results for the postulated flaw under 60 °F/h and 100 °F/h rates of cool-down and heat-up P-T limit transients, respectively. The figures illustrate the a_{KIC} and applied- K_I variations during the P-T limit transients. From Fig. 8, it is seen that the applied- K_I of cool-down P-T limit transients rises gradually ($dK_I/dt > 0$), so the peak or the final maximum value of applied- K_I is crucial in determination of CPI according to the WPS model. Therefore, the higher rate of cool-down transient results in the higher value of fracture probability. On the contrary, the CPI for heat-up transients is determined by the maximum value of applied- K_I at starting point since the applied- K_I during heat-up transients is decreasing. Consequently, the fracture probabilities for 60 °F/h rate of heat-up P-T limit transients are higher than those for 100 °F/h rate of heat-up P-T limit transients.

- (2) On the other hand, Figs. 8 and 9 also explain the reason that the fracture probabilities for the pressurized heat-up transients are lower than the depressurized cool-down transients, due to the decreasing applied- K_I associated with the increasing a_{KIC} and the increasing applied- K_I associated with the decreasing a_{KIC} under heat-up and cool-down transients, respectively.
- (3) For some cool-down P-T limit transients at lower embrittlement levels, the fracture probabilities analyzed from the K_{Ia} -based limits are higher than those from the K_{IC} -based limits which have higher permissible pressure. Fig. 10 shows the deterministic analysis results for 20 °F/h rate of cool-down P-T limit transients at various embrittlement levels. For 36 EFPY and 54 EFPY conditions, it can be seen that although the applied- K_I of K_{IC} -based limits are higher than those of the K_{Ia} -based limits, they finally do not exceed the previous peak values after the applied- K_I are larger than $a_{KIC}(\tau)$. Therefore, during the entire transient the flaw is in the state of WPS, leading to the exclusion of crack initiation. In contrast, the applied- K_I of both K_{Ia} -based limits and K_{IC} -based limits with higher embrittlement finally rise to exceed their previous peak values. Consequently, the flaw leaves the WPS state and the P-T limits which produce higher applied- K_I lead to a higher CPI.

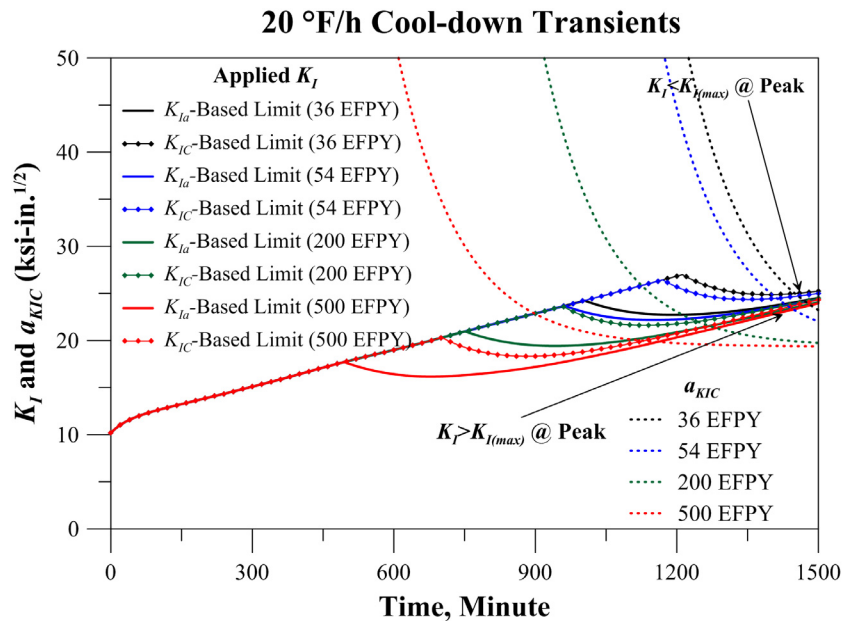


Fig. 10. Comparison of the deterministic analysis results for the cool-down P-T limit transients at various embrittlement levels.

7. Conclusion

This paper compares various ASME pressure-temperature limits on the structural integrity of a Taiwan domestic PWR pressure vessel. The probabilistic approach has been conducted by employing PFM FAVOR code associate with the flaw characteristics generated by the VFLAW code for the domestic PWR pressure vessel. It is seen that the fracture probability of RPV under P-T limit transient increases significantly with the increased radiation embrittlement, and the cladding-induced shallow surface breaking flaws dominate the fracture behavior. The analyzed results indicate that the P-T curves based on both the K_{IC} curve proposed by the licensee and the risk-informed methodology which provides more operational flexibility can meet the safety goal of $CPF < 1 \times 10^{-5}$ until no less than the proposed extended operation period of the RPV (54 EFPY).

On the other hand, the deterministic analyses have also been performed to examine the discrepancies from the PFM analysis results. The illustrations of applied- K_I and fracture toughness a_{KIC} variation indicate that the warm-prestressing significantly affects the fracture behavior of a flaw during P-T limit transients. If the flaws are in the state of warm-prestressing, the crack initiation will be excluded even with higher applied- K_I than a_{KIC} , and thus reduce the fracture probability.

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